

Mid Semester Examination March 2025

Name of the Program: B. Tech (EE)
Name of the Course: Power System Protection

Semester: VI
Course Code: TEE 603

Time: 90 minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
(ii) Each question carries 10 marks

Q1	(10 marks)						CO 1
(a)	Discuss the term 'Switchgear'. Why time delay is intentionally provided in practical relays?						
OR							
(b)	Explain the basic essential qualities of a protective scheme.						
Q2	(10 marks)						CO 1
(a)	Analyze the behavior of a transmission line under sudden occurrence of fault and discuss doubling effect.						
OR							
(b)	A transmission of inductance 0.1 H and resistance 5 ohm is suddenly short-circuited at the far end. Write the expression for short circuit current $i(t)$. Consider $v = 100\sin(100\pi t + 15^\circ)$						
Q3	(10 marks)						CO 2
(a)	Describe the construction and working of a Buchholz relay.						
OR							
(b)	With a neat diagram, explain zones of protection in a power system.						
Q4	(10 marks)						CO 2
(a)	Define PSM. How are relays classified on the basis to the functions in the protective scheme?						
OR							
(b)	The current setting is set at 150%, TMS = 0.4; C.T. Ratio = 400/5, fault current is 6000 A. Determine the operating time of the relay. At TMS = 1 the operating time at various PSMs are shown in the table below:						
	PSM	2	4	5	8	10	20
	Operating time in secs.	10	5	4	3	2.8	2.4
Q5	(10 marks)						CO 3
(a)	Describe the principle of impedance type distance relays.						
OR							
(b)	What is the principle of a differential relay? What are its drawbacks and how are they overcome? Explain.						

